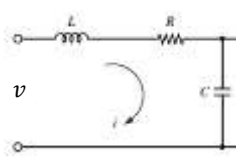
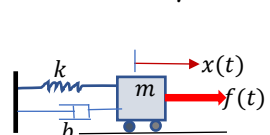
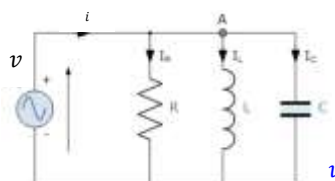
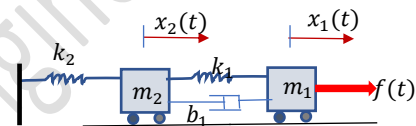


Electrical-Mechanical Analogous systems

Series $R - L - C$ circuit	Mechanical System	Parallel $R - L - C$ circuit
 $i = \frac{dq}{dt}$ $L \frac{di}{dt} + Ri + \frac{1}{C} \int i dt = v$ $L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = v$	 $m\ddot{x} + b\dot{x} + kx = f(t)$ $m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + kx = f(t)$	 $v = \frac{d\phi}{dt}$ $\frac{v}{R} + C \frac{dv}{dt} + \frac{1}{L} \int v dt = i$ $C \frac{d^2\phi}{dt^2} + \frac{1}{R} \frac{d\phi}{dt} + \frac{\phi}{L} = i$
Voltage Charge Current Inductance Resistance $\frac{1}{\text{Capacitance}}$	Force Displacement Velocity Mass Damping coefficient Spring constant	Current Flux Voltage Capacitance $1/\text{Resistance}$ $1/\text{Inductance}$

Example 1

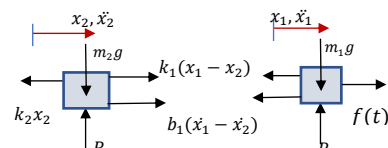
Obtain the force-voltage and force-current analogous systems for the mechanical system shown.



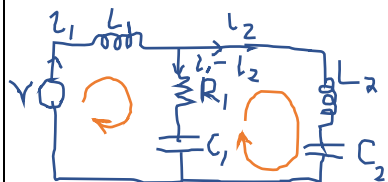
The governing differential equations for the mechanical system are

$$m_1 \ddot{x}_1 + b_1(\dot{x}_1 - \dot{x}_2) + k_1(x_1 - x_2) = f(t)$$

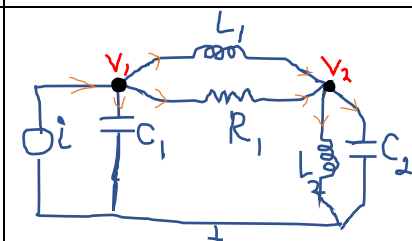
$$m_2 \ddot{x}_2 - b_1(\dot{x}_1 - \dot{x}_2) - k_1(x_1 - x_2) + k_2 x_2 = 0$$



$F \rightarrow v$ $x \rightarrow q$ $\dot{x} \rightarrow i$ $m \rightarrow L$ $b \rightarrow R$ $k \rightarrow \frac{1}{C}$	$L_1 \ddot{q}_1 + R_1(\dot{q}_1 - \dot{q}_2) + \frac{1}{C_1}(q_1 - q_2) = v$ $L_1 \frac{di_1}{dt} + R_1(i_1 - i_2) + \frac{1}{C_1} \int (i_1 - i_2) dt = v$ $L_2 \ddot{q}_2 - R_1(\dot{q}_1 - \dot{q}_2) - \frac{1}{C_1}(q_1 - q_2) - \frac{1}{C_2} q_2 = 0$ $L_2 \frac{di_2}{dt} - R_1(i_1 - i_2) - \frac{1}{C_1} \int (i_1 - i_2) dt + \frac{1}{C_2} \int i_2 dt = 0$
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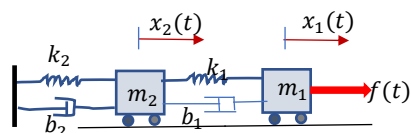


$F \rightarrow i$ $x \rightarrow \phi$ $\dot{x} \rightarrow v$ $m \rightarrow C$ $b \rightarrow \frac{1}{R}$ $k \rightarrow \frac{1}{L}$	$C_1 \ddot{\phi}_1 + \frac{1}{R_1}(\dot{\phi}_1 - \dot{\phi}_2) + \frac{1}{L_1}(\phi_1 - \phi_2) = i$ $C_1 \frac{dv_1}{dt} + \frac{1}{R_1}(v_1 - v_2) + \frac{1}{L_1} \int (v_1 - v_2) dt = i$ $C_2 \ddot{\phi}_2 - \frac{1}{R_1}(\dot{\phi}_1 - \dot{\phi}_2) - \frac{1}{L_1}(\phi_1 - \phi_2) + \frac{1}{L_2} \phi_2 = 0$ $C_2 \frac{dv_2}{dt} + \frac{1}{L_2} \int v_2 dt = \frac{1}{R_1}(v_1 - v_2) + \frac{1}{L_1} \int (v_1 - v_2) dt$
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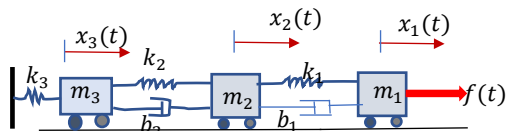


Exercise 1

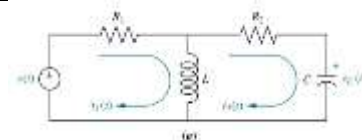
Obtain the force-voltage and force-current analogous systems for the mechanical system shown.

**Exercise 2**

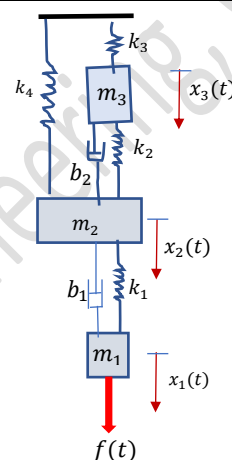
Obtain the force-voltage and force-current analogous systems for the mechanical system shown.

**Exercise 3**

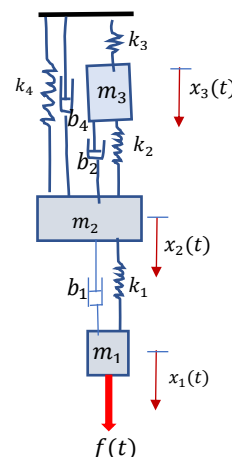
Obtain the equivalent mechanical system

**Exercise 4**

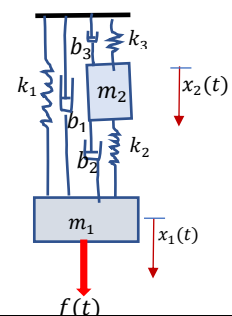
Obtain the force-voltage and force-current analogous systems for the mechanical system shown.

**Exercise 5**

Obtain the force-voltage and force-current analogous systems for the mechanical system shown.

**Exercise 6**

Obtain the force-voltage and force-current analogous systems for the mechanical system shown.



References

1. Modern Control Systems : Richard C. Dorf and Robert H. Bishop, 8/e, Addison-Wesley
2. Modern Control Engineering : Katsuhiko Ogatta, Pearson Education, LPE
3. Control Systems Engineering: Norman S. Nise, 6/e, John Wiley & Sons